NETWORKING

Anycast Visual Guide

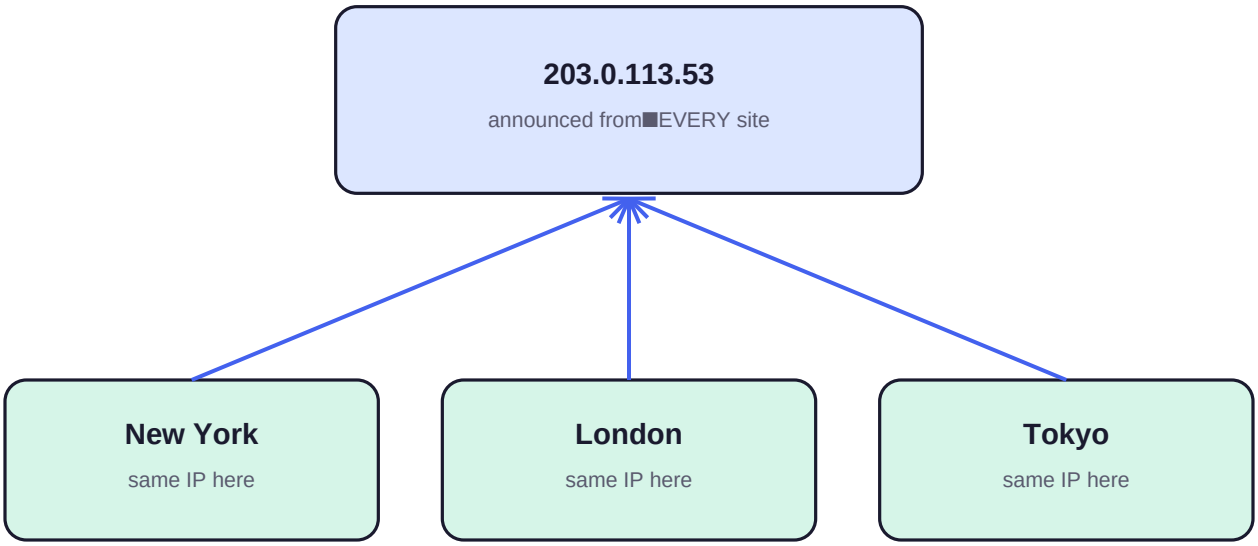
One IP, many places —
how the internet finds the nearest you.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

One IP Announced Everywhere

The same prefix advertised from dozens of locations at once.

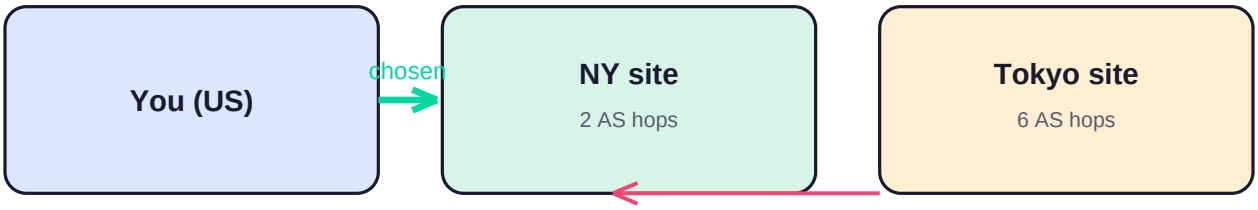


BGP sees multiple equal origins for one prefix — and picks the nearest per viewer.

STEP 2/5

BGP Nearest-Wins

Each network's routers choose the closest announcement.



- 'Nearest' = fewest AS hops by BGP policy — usually, but not always, geographic.
- No client configuration: the routing system makes the decision transparently.
- Return path may differ (asymmetric routing) — design stateless services accordingly.

STEP 3/5

DNS Roots & CDNs Run on Anycast

The internet's most critical services share this trick.

DNS root servers

13 logical IPs, 1000+ physical instances — a root is always nearby.

Authoritative DNS

Cloudflare, Route53, NS1 — anycasted DNS survives regional outages.

CDNs

One IP per edge PoP cluster — users land on the closest cache.

DDoS absorption

Attack traffic splits across sites instead of drowning one.

STEP 4/5

Unicast vs Anycast vs Multicast

Three ways to address 'them'.

Unicast

one → one. A specific host. The default.

Anycast

one → nearest-of-many. Same IP, many sites.

Multicast

one → many-subscribers. IPTV, OSPF hellos (224.0.0.0/4).

Broadcast

one → everyone on the LAN. Dies at the router.

STEP 5/5

Anycast Gotchas

Nearest isn't always stable — plan for flapping.

- Route flaps can move a client mid-session to a different site — keep services stateless or share session state.
- TCP + anycast: generally fine while routes are stable; pathological flapping breaks long connections.
- Debugging is harder: 'which site answered?' needs per-site identifiers in responses.
- Not for stateful backends: anycast the stateless edge, unicast the databases behind it.

Anycast + health checks: withdraw the announcement from a sick site and traffic drains away — failover without DNS TTL waits.

STEP ✓/✓

Key Takeaways

Anycast — Remember

- Anycast = one IP announced from many sites; BGP routes each client to the nearest.
- Zero client config — the routing system decides transparently.
- Powers DNS roots, authoritative DNS, CDNs, and DDoS absorption.
- Unicast = one host; anycast = nearest of many; multicast = subscribers; broadcast = LAN-only.
- Keep anycasted services stateless; withdraw sick sites via BGP for instant failover.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev